

AUTOMATION PORTFOLIO — TECHNICAL RECORD

SOPHON FINANCE SYSTEMS

AI-Driven Finance & Accounting Automation | github.com/sophonfinance-wq/finance-automation-portfolio

KNOWLEDGE BRAIN

A deterministic, citation-governed knowledge base that turns fictional finance/tax meeting transcripts into verbatim, timestamped, source-cited answers, and refuses to answer when it has no source.

Report No.	SFS-ENG-007	Package	brain_engine
System	Knowledge Brain	Data	Fictional / seeded
Run	python -m brain_engine ask "how is return of capital beyond basis treated?"	Status	Public — portfolio record

1.0 OVERVIEW AND SCENARIO

During the seeded fictional review meeting "Surplus Workpaper Review — Reviewer Corrections" (2025-04-08), reviewer Quinn Harlow dictates four spoken corrections — for example, "Change the distribution formula to reference column E, not column D, across every year column." Months later that instruction would normally survive only as a vague memory. The engine captures each correction verbatim as a cited ChangeDirective and auto-writes a paste-ready remediation prompt plus a 1:1 change-log, so the fix is applied word-for-word rather than paraphrased from recall. Its integrity loop separately catches knowledge-base drift injected in the demo — a card's rule text quietly edited after storage, a speaker mis-stamped (Priya Vale re-stamped "A. Nobody"), and an orphaned citation pointing at 03:59:59 where no one spoke — re-deriving the repairable entries verbatim and refusing to invent a source for the orphan.

2.0 WHAT THE SYSTEM AUTOMATES

- Parses fully fictional meeting transcripts into timestamped utterances, rendering HH:MM:SS from integer second offsets
- Extracts flagged authoritative utterances into knowledge cards (decision / rule / definition / open-item), each carrying full provenance and text byte-identical to the source
- Builds a deterministic keyword/tag TF-IDF index over the cards using the Python standard library only (no embeddings, no LLM, no network)
- Answers three query shapes: ask (top matching cited cards), cite (a single paste-ready verbatim footnote), and prep (settled positions plus parked open items)
- Captures a review meeting's spoken corrections as ChangeDirectives and generates an apply-ready remediation prompt plus a cited change-log at status PENDING
- Writes five Markdown deliverables (brain index, citation example, meeting-prep example, remediation prompt, change log) via the --out flag
- Runs an integrity loop that re-verifies every stored card and directive against the transcript of record, re-derives drift verbatim, and quarantines uncited (orphaned) entries

- Enforces refusal governance: returns an explicit refusal with a scriptable exit code when nothing clears the relevance floor

3.0 OPERATING PROCEDURE

- 1 `build_corpus()` hand-builds the seeded, deterministic fictional corpus of 5 meetings, each a tuple of timestamped Utterances (`DEFAULT_SEED = 1107`; the seed is accepted only for API parity).
- 2 Extraction lifts flagged utterances into 15 KnowledgeCards and 4 ChangeDirectives, copying the spoken text byte-for-byte and stamping Provenance (meeting id, title, date, speaker, HH:MM:SS).
- 3 BrainEngine builds a TF-IDF index once at construction: tokenize (lowercase, drop stopwords, preserve hyphenated compounds), weight topic tags 3x, then compute smoothed IDF and cosine-normalized per-card vectors.
- 4 `rank()` scores every card by cosine similarity to the query, discards anything below `MIN_RELEVANCE` (0.05), and sorts by descending score with `card_id` as a deterministic tiebreak.
- 5 `ask / cite / prep` consume that ranking: `cite` filters to authoritative kinds and returns the single best verbatim footnote, `prep` retains open-items and orders them last, and an empty result triggers the refusal banner.
- 6 `remediate()` resolves a review title or topic to its directives in spoken (meeting, timestamp) order and renders a ready-to-paste prompt plus a change-log mapping each directive 1:1 to its source citation at status `PENDING`; no match yields an empty, refusing packet.
- 7 `report.py` renders the Markdown; the CLI `--out` flag writes the five deliverables and returns exit code 0 on a sourced answer or 3 on refusal.
- 8 `loop.py` re-verifies each stored entry against `Corpus.utterance_text`, re-derives `TAMPERED_TEXT` and `TAMPERED_SPEAKER` findings verbatim, quarantines `ORPHANED` findings, and returns `CLEAN / REMEDIATED / ESCALATED` (exit 0 / 0 / 3).

4.0 CONTROLS AND SAFEGUARDS

- Citation-governed retrieval: every returned answer carries full provenance (meeting, date, HH:MM:SS, speaker); no card exists without a source, and this is asserted both in the model and the test suite
- Refuse-do-not-guess: any card scoring below the 0.05 relevance floor is excluded, so off-topic queries return an explicit refusal with scriptable exit code 3 rather than a fabricated answer
- Verbatim provenance integrity: card `rule_text` and directive `request_text` are copied byte-for-byte from the source utterance, and `cite` mode returns text byte-identical to the transcript
- Determinism: a fixed corpus yields identical TF-IDF scores and ranked order every run, with `card_id` tiebreaks giving total ordering; computation is `stdlib`-only with no network, LLM, or embeddings
- Read-only engine: BrainEngine never mutates the corpus, and the index is built once at construction
- Authority gating and integrity loop: open-items surface in `prep` briefings but are never returned as workpaper authority, and the loop re-derives KB drift verbatim while quarantining orphaned citations (no source, no guess) and escalating with the refusal exit code

5.0 VERIFICATION AND TESTING

Correctness is proven by an automated pytest suite that passes in full (1,511 tests, verified by running pytest in the clone; the README badge cites 1,499). The suite asserts the load-bearing governance: every returned answer carries a citation, refuse-do-not-guess when nothing clears the relevance floor, HH:MM:SS

formatting from second offsets, cite-mode text byte-identical to the source, retrieval determinism (same query yields the same ranked order and scores), prep inclusion/exclusion of the right cards, index counts, and that no card exists without a source; the review-to-remediation tests add full-provenance directive extraction, a prompt containing every directive in order with verbatim quotes, a change-log mapping 1:1 at status PENDING, prompt determinism, and refusal when a review topic has no directives. The integrity loop is separately demonstrable: ``python -m brain_engine.loop --demo`` injects four drift faults, re-derives three verbatim, quarantines one orphan, and exits with refusal code 3.

6.0 KEY FACTS

Attribute	Value
Meetings ingested	5
Knowledge cards	15 (4 decisions, 4 rules, 3 definitions, 4 open-items)
Change-directives captured	4 (from the review meeting)
Relevance floor (MIN_RELEVANCE)	0.05
Automated tests passing	1,511 (verified via pytest)
Refusal exit code	3

This record documents a self-contained system in the Sophon Finance Systems automation portfolio. All data is fictional and seeded; no employer or client workpaper, entity, methodology, path, or figure is reproduced. Prepared 14 July 2026.